Pakistan Cotton Supply and Demand: 2026/27 Scenario Analysis

Nimra Shahab, November 2026. Data source: USDA PSD Online.

Background

Pakistan's cotton production was about 8.2 million bales in 2017/18, its highest in the data I used. USDA's forecast for 2026/27 is about 5 million, which is roughly 39% lower. Harvested area also fell, from 2.9 million hectares in 2015/16 to about 1.9 million. Mills still use around 10 million bales a year, so the gap is filled by imports. Imports were about 31% of mill use in 2017/18. For 2026/27 they are forecast at 5.1 million bales, about half of mill use.

What I did

I downloaded 12 years of USDA data and checked it before using it. I tested that the supply and distribution totals add up, that each year's ending stocks match the next year's beginning stocks, and that yield matches production and area. All checks passed. I then built a scenario model on the 2026/27 forecast. I chose the size of the yield changes from how much yield has varied in this data, and compared them with real cases such as the 2022 floods and the 2024 heat and pest losses. I also compared USDA production and area with the Pakistan Economic Survey for 2015/16 to 2023/24.

Results

Scenario	Yield change	Production (1,000 bales)	Imports needed (1,000 bales)
Base (USDA forecast)	0%	5,000	5,100
Bad weather	-15%	4,250	5,850
Pest outbreak	-10%	4,500	5,600
Severe (heat and floods, area -5%)	-20%	3,800	6,300
Good season	+10%	5,500	4,600

A bad year adds 500 to 1,200 thousand bales of imports, while a good year saves only about 500, so the downside is larger than the upside. Using my own probabilities for each scenario, expected imports are about 5,400 thousand bales, compared with 5,100 in the base case. Figures are rounded.

In the comparison with the Economic Survey, the two sources move together (correlation about 0.97), but USDA's production is lower in 8 of the 9 years, usually by about 10%. This means my levels may be on the low side, while the trends and the scenario results should still hold.

Limits

- Mill use, exports and ending stocks are kept fixed. Prices and policy are not modelled.
- The yield changes and the probabilities are my own judgment, not forecasts.
- The latest year is a USDA forecast and will be revised.
- The Economic Survey comparison stops at 2023/24. Checking 2024/25 onward is still to do.

Sources

USDA Foreign Agricultural Service, PSD Online (Cotton, Pakistan), downloaded November 2026. Pakistan Economic Survey 2023-24, Chapter 2 Agriculture. Pakistan Economic Survey 2022-23 and press reports on the 2022 floods and the 2024 season. The full workings are in the shared spreadsheet and the GitHub repository.